

METIMMOX-1 Cost-Effectiveness Analysis Report

Biomarker-Guided Strategies for Selecting Immunotherapy Candidates in Metastatic MSS/pMMR Colorectal Cancer

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1 Overview

This cost-effectiveness analysis evaluates three biomarker-guided strategies (CRP, TLR, and TMB/BRAF) for selecting metastatic microsatellite stable (MSS)/proficient mismatch repair (pMMR) colorectal cancer patients who might benefit from immunotherapy with nivolumab plus FLOX versus standard FLOX chemotherapy.

This report contains:

- **Strategy Descriptions:** Overview of the four treatment strategies evaluated
- **Base Case Analysis:** Cost-effectiveness results with incremental cost-effectiveness ratios (ICERs) and cost-effectiveness plane
- **Deterministic Sensitivity Analysis:** Tornado diagrams showing parameter impact on each strategy
- **Probabilistic Sensitivity Analysis:** Monte Carlo simulation results including scatter-plot and cost-effectiveness acceptability curves

2 Strategies Evaluated

Table 1: Treatment Strategies Evaluated in the Cost-Effectiveness Analysis

Strategy	Description	Biomarker Prevalence	Patients (n)
Standard of Care	Standard of care - All patients receive only FLOX chemotherapy	100.0%	36
CRP-guided Strategy	C-reactive protein with cut-off of <5 for biomarker-positive status. If CRP-positive: alternating two cycles each of FLOX (chemotherapy) and nivolumab (anti-PD1 immunotherapy); if CRP-negative: chemotherapy only	33.8%	45
TLR-guided Strategy	Tumor Lesion Reduction with cut-off of $\geq 10\%$ for biomarker-positive status. If TLR-positive: alternating two cycles each of FLOX (chemotherapy) and nivolumab (anti-PD1 immunotherapy); if TLR-negative: chemotherapy only	64.7%	26
TMB/BRAF-guided Strategy	Combined biomarker: either Tumor Mutation Burden ≥ 9 or BRAF V600 mutation positive (both from next-generation sequencing). If TMB/BRAF-positive: alternating two cycles each of FLOX (chemotherapy) and nivolumab (anti-PD1 immunotherapy); if TMB/BRAF-negative: chemotherapy only	44.9%	34

3 Base Case Analysis

Table 2: Base Case Cost-Effectiveness Results with ICERs

Strategy	Cost	Effect	Inc_Cost	Inc_Effect	ICER	Status
control	\$20,894	1.844	NA	NA	NA	ND
tmb_braf	\$63,560	1.870	42665.89	0.0252389	\$1,690,480	ND
crp	\$54,898	1.784	NA	NA	NA	D
tlr	\$84,974	1.704	NA	NA	NA	D

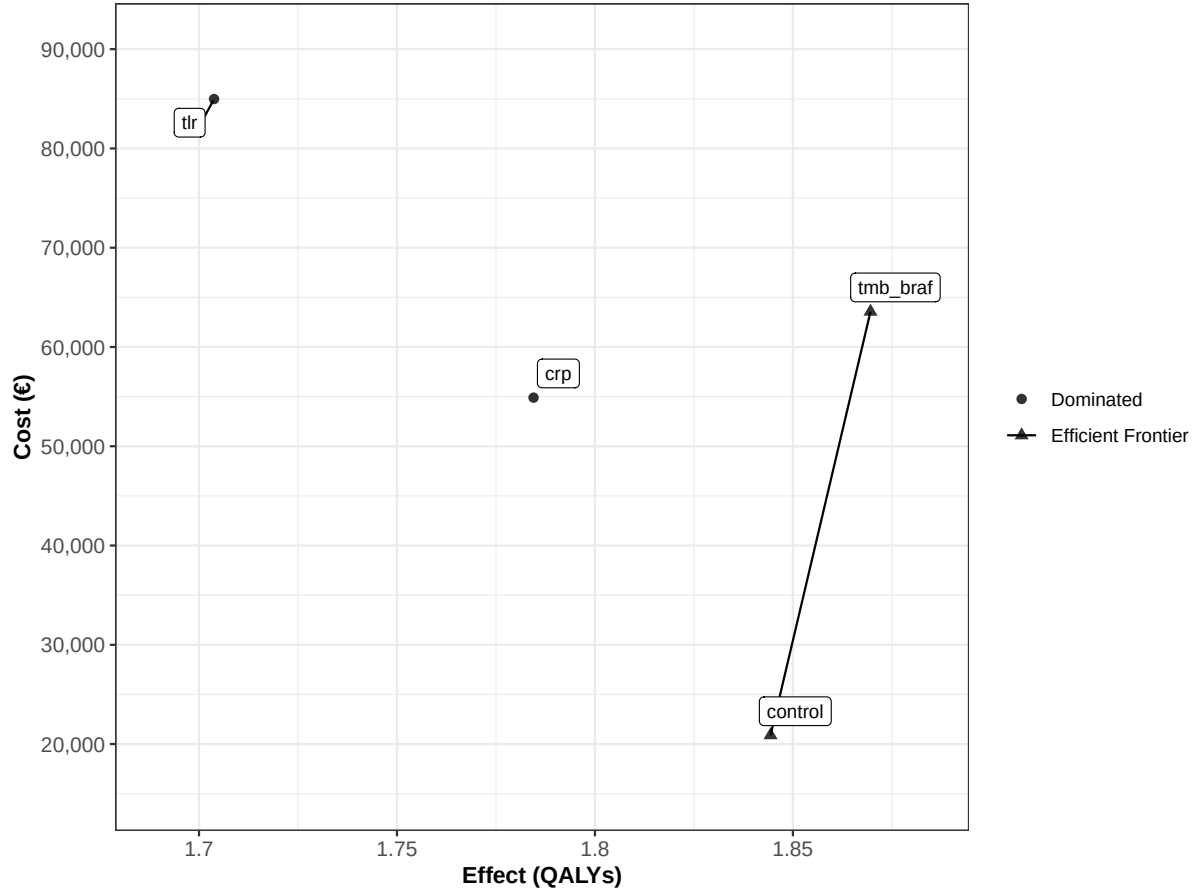


Figure 1: Cost-Effectiveness Plane with Efficiency Frontier

4 Deterministic Sensitivity Analysis

To assess the robustness of the base case results, we conducted one-way sensitivity analyses for each of the four treatment strategies. The tornado diagrams below show the impact of varying individual model parameters within their plausible ranges on the net monetary benefit (NMB) for each strategy.

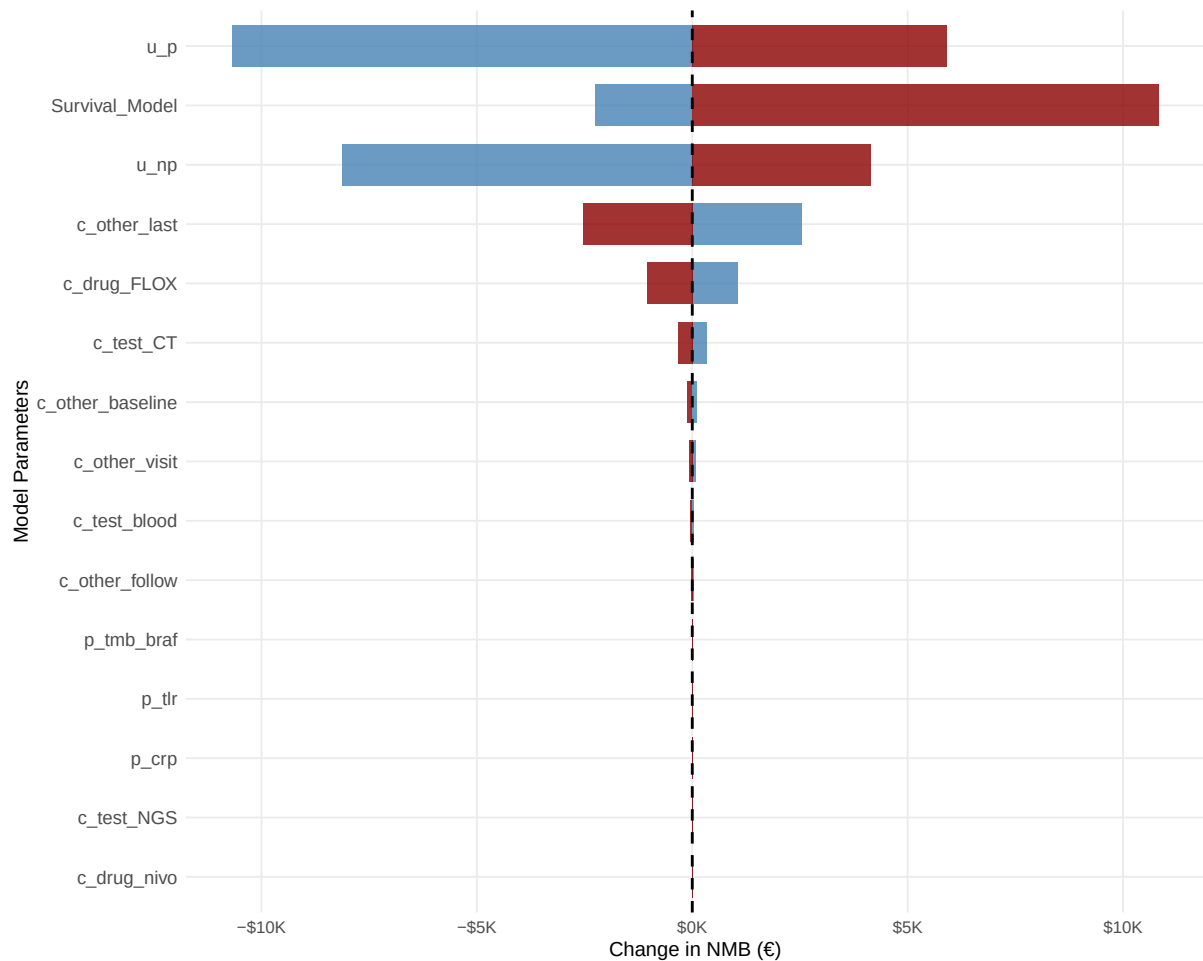


Figure 2: Tornado Diagram - Standard of Care One-Way Sensitivity Analysis

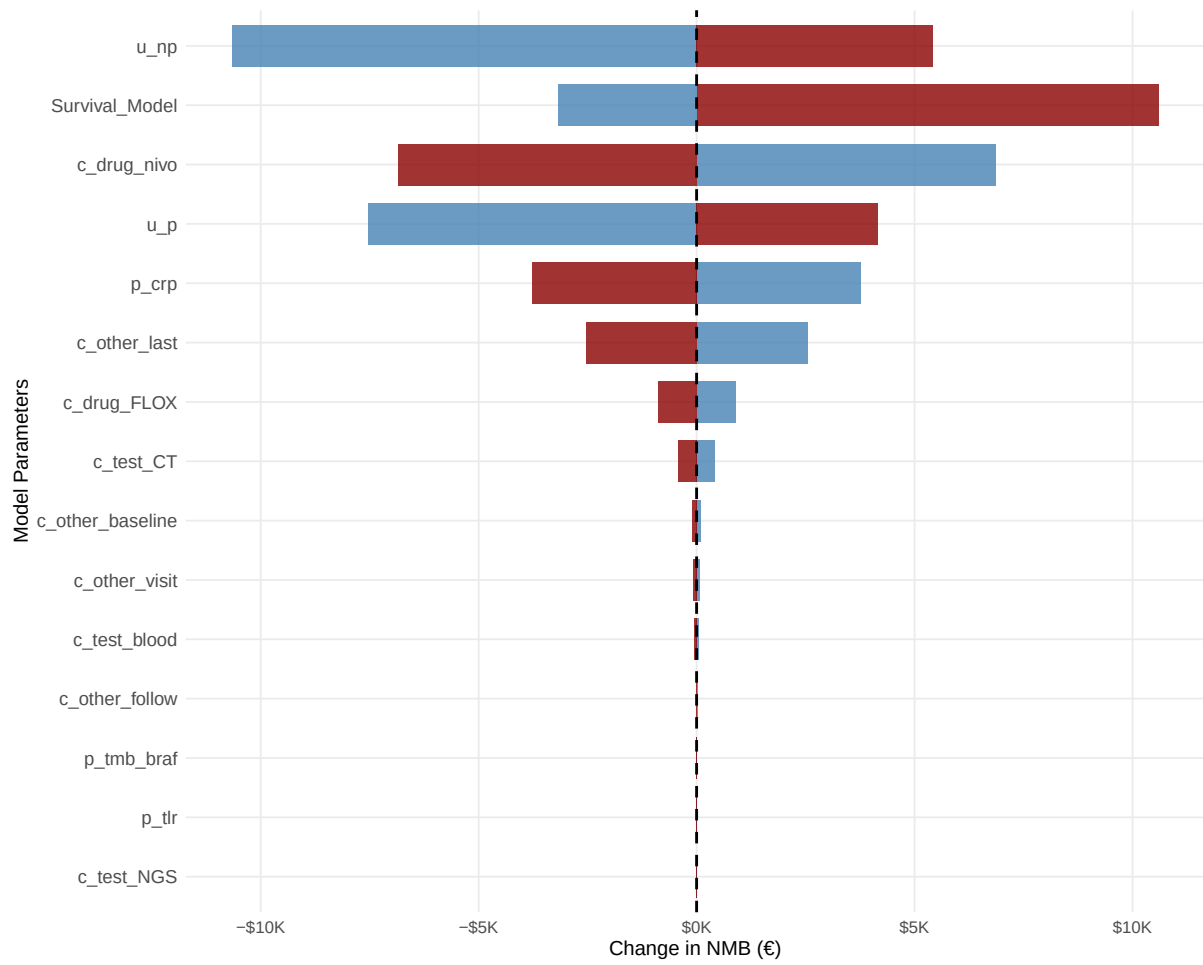


Figure 3: Tornado Diagram - CRP-guided Strategy One-Way Sensitivity Analysis

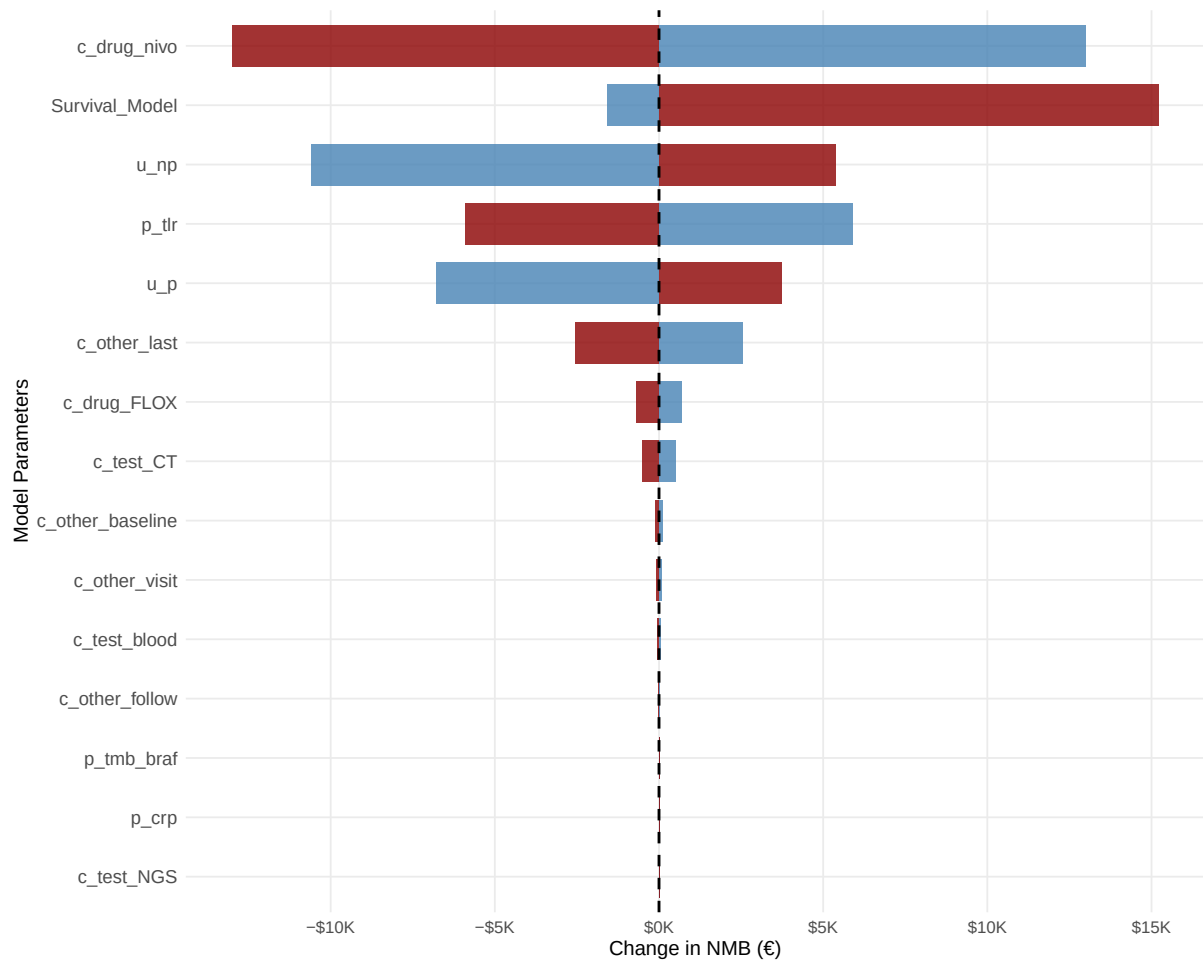


Figure 4: Tornado Diagram - TLR-guided Strategy One-Way Sensitivity Analysis

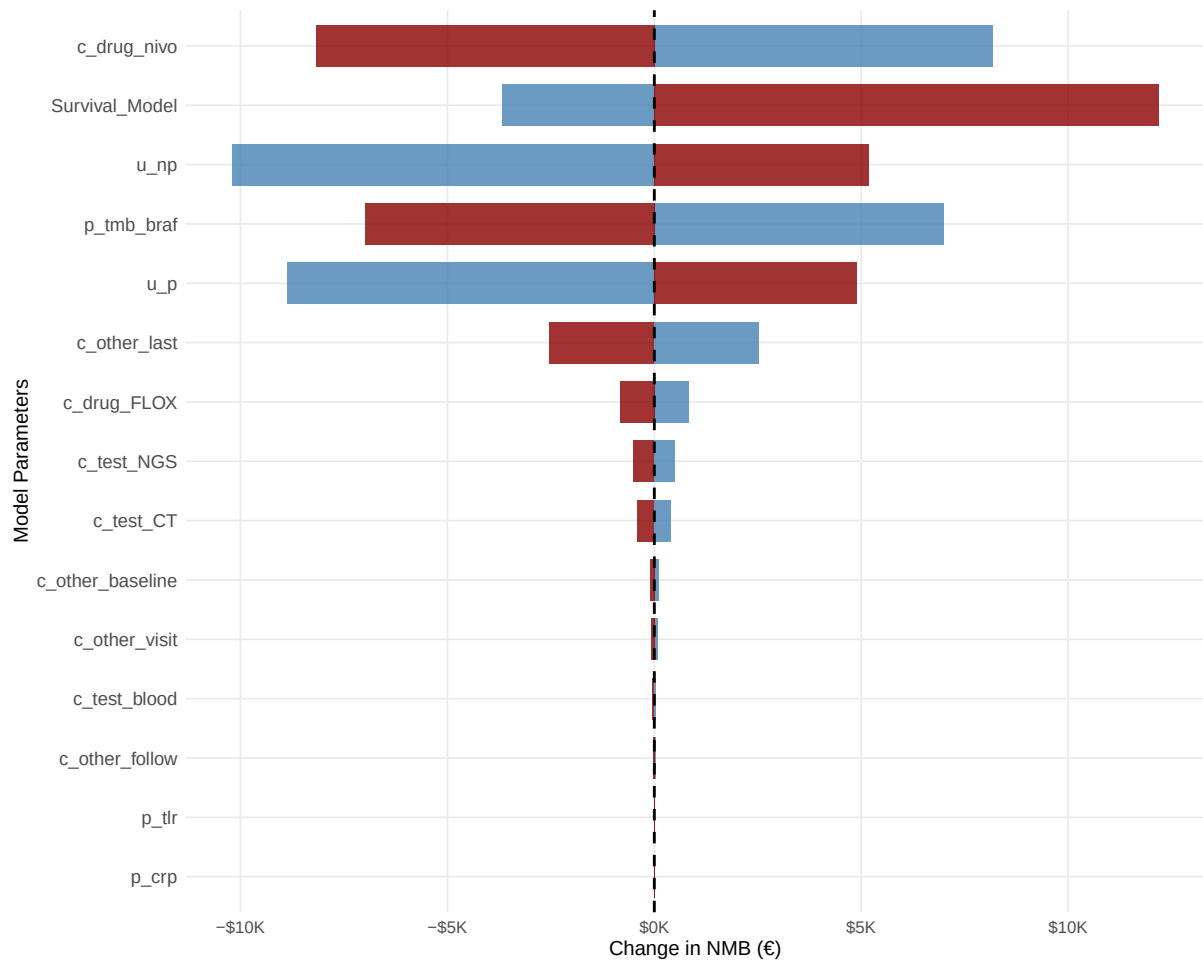


Figure 5: Tornado Diagram - TMB/BRAF-guided Strategy One-Way Sensitivity Analysis

5 Probabilistic Sensitivity Analysis

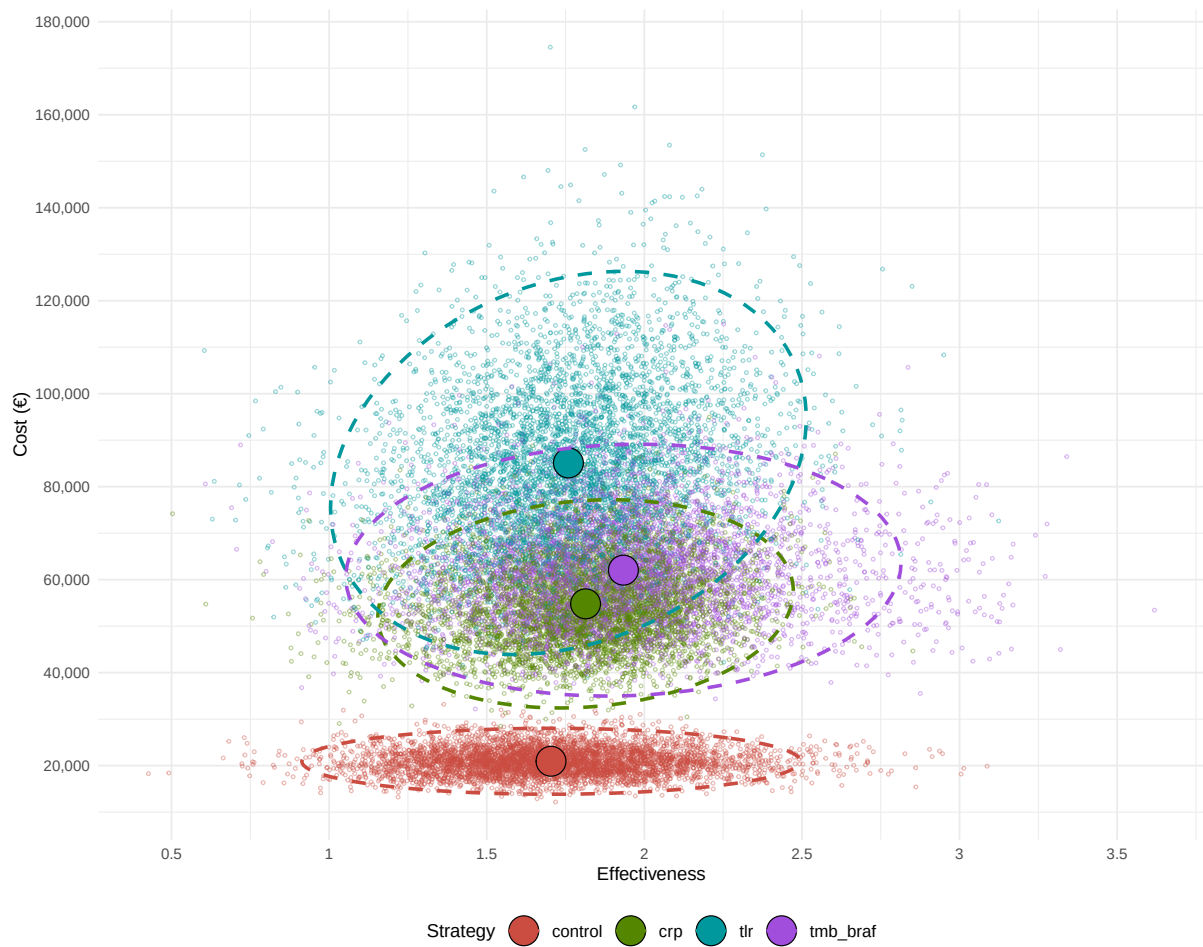


Figure 6: Cost-effectiveness Scatterplot from Probabilistic Sensitivity Analysis

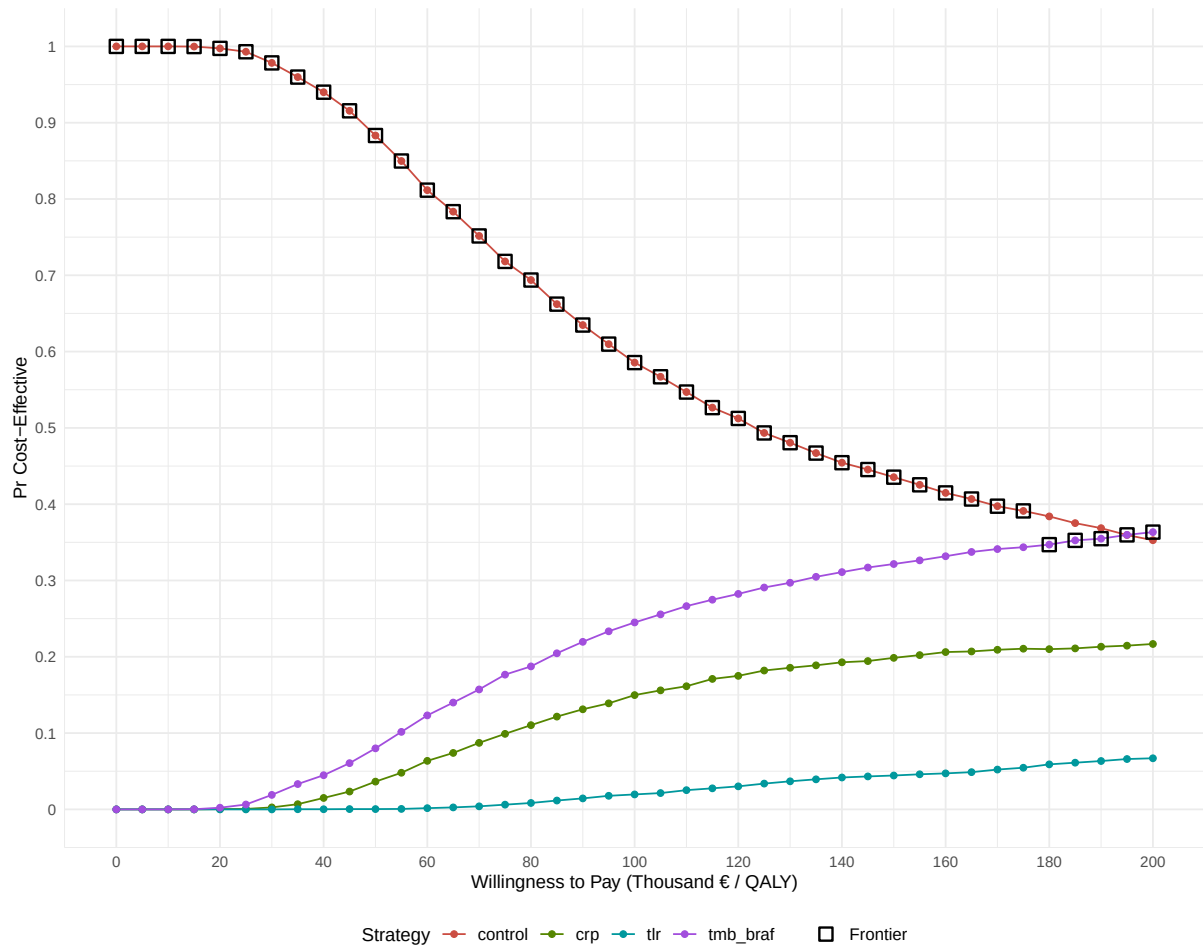


Figure 7: Cost-Effectiveness Acceptability Curve (CEAC)

6 Other Reports

I am working on the following other reports:

- **Input Parameters Report:** Comprehensive documentation of all model input parameters including clinical effectiveness data, cost parameters, and utility values with their respective sources and uncertainty distributions.
- **Parametric Survival Analysis Report:** Detailed documentation of the parametric survival modeling process, including goodness-of-fit statistics for all tested distributions, model selection criteria, and validation against Kaplan-Meier estimates.
- **One-Way Sensitivity Analysis (OWSA) Report:** Extended sensitivity analysis providing detailed examination of each individual parameter's impact on model outcomes, including threshold analyses for key decision-relevant parameters.

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Repository: ben-geisler/METIMMOX-1

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